

Modified Equation Analysis

Idea: Given a discretization, find a modified PDE such that the discrete solution is the exact solution of the modified PDE.

Example: $u_t + au_x = 0$

Forward time, centered space:

$$\frac{U_j^{n+1} - U_j^n}{K} + \frac{a}{2h}(U_{j+1}^n - U_{j-1}^n) = 0$$

Suppose there exists $V(x,t)$ such that if we replace $U_j^n \rightarrow V(x_j, t_n)$, the equation above is exactly satisfied.

$$\frac{V(x_j, t_n + k) - V(x_j, t_n)}{k} + \frac{a}{2h} (V(x_{j+1}, t_n) - V(x_{j-1}, t_n)) = 0$$

$$\frac{kV_t(x_j, t_n) + \frac{k^2}{2}V_{tt}(x_j, t_n) + O(k^3)}{k} + \frac{a}{2h} \left(2hV_x + 2\frac{h^3}{6}V_{xxx} + O(h^5) \right) = 0$$

$$V_t + aV_x = -\frac{k}{2}V_{tt} + O(k^2, h^2)$$

$$\Rightarrow V_t + aV_x = -\frac{k}{2}a^2V_{xx} + O(k^2, h^2)$$

anti-diffusive
(blowup)

$$V_t + aV_x = O(k, h^2)$$

$$V_t - aV_x = O(k, h^2)$$

$$V_{tt} = -aV_{xt} + O(\dots)$$

$$V_{tx} = -aV_{xx} + O(\dots)$$

$$V_{tt} = a^2V_{xx} + O(\dots)$$

Lax-Friedrichs

$$U_j^{n+1} = \frac{1}{2}(U_{j+1}^n + U_{j-1}^n) - \frac{Ka}{2h}(U_{j+1}^n - U_{j-1}^n)$$

$$U_j^n \rightarrow V(x_j, t_n)$$

$$V(x_j, t_n + k) = \frac{1}{2}(V(x_j + h, t_n) + V(x_j - h, t_n)) - \frac{Ka}{2h}(V(x_j + h, t_n) - V(x_j - h, t_n))$$

$$V_t + aV_x = -\frac{k}{2}V_{tt} + \frac{h^2}{2k}V_{xx} + \mathcal{O}(k^2, h^2)$$

$$V_t + aV_x = \left(\frac{h^2}{2k} - \frac{Ka^2}{2}\right)V_{xx} + \mathcal{O}(k^2, h^2)$$

must be
positive

$$\frac{h^2}{2k} - \frac{Ka^2}{2} \geq 0 \Rightarrow h^2 \geq Ka^2$$

$$\left| \frac{Ka}{h} \right| \leq 1$$

Condition for
stability.

$$\frac{Ka}{h} = \nu$$

Courant number
or

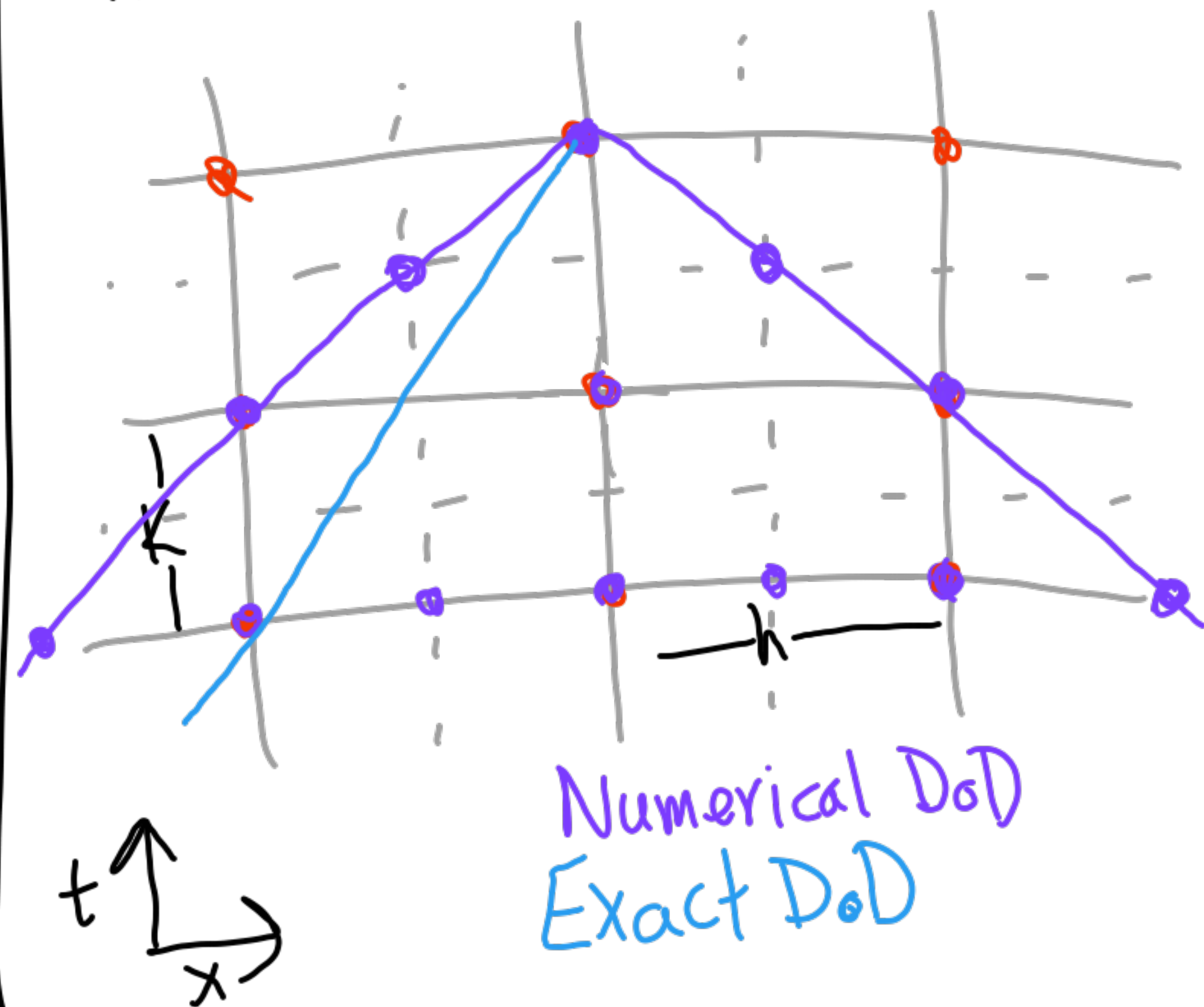
CFL number

CFL condition

(Courant, Friedrichs, Lewy 1927)

The domain of dependence (DoD) of the exact solution of a PDE must be contained in the DoD of the numerical solution, as $K, h \rightarrow 0$, for a method to be convergent.

The Domain of Dependence: for a point (x^*, t^*) is the set of all points (x, t) that can influence $u(x^*, t^*)$.



The CFL condition is necessary but not always sufficient for convergence.

— here CFL is satisfied if $|ka| \leq h$.

Centered in time + space
(Leapfrog)

$$\frac{U_j^{n+1} - U_j^{n-1}}{2k} + a \frac{U_{j+1}^n - U_{j-1}^n}{2h} = 0$$

$$\frac{V(x_j, t_n + k) - V(x_j, t_n - k)}{2k} + a \frac{V(x_j + h, t_n) - V(x_j - h, t_n)}{2h} = 0$$

$$V_t + \frac{k^2}{6} V_{ttt} + \mathcal{O}(k^4) + a \left(V_x + \frac{h^2}{6} V_{xxx} + \mathcal{O}(h^4) \right) = 0$$

$$V_t + aV_x = -\frac{k^2}{6} V_{ttt} - a\frac{h^2}{6} V_{xxx} + \mathcal{O}(k^4, h^4)$$

$$V_t + aV_x = \frac{a}{6} (a^2 k^2 - h^2) V_{xxx} + \mathcal{O}(k^4, h^4, k^2 h^2)$$

$$V_t = -aV_x + \mathcal{O}(k^2, h^2)$$

$$V_{tt} \approx -aV_{xt}$$

$$V_{tx} \approx -aV_{xx}$$

$$V_{tt} \approx a^2 V_{xx}$$

$$V_{ttt} \approx a^2 V_{xxt}$$

$$V_{txx} \approx -aV_{xxx}$$

$$V_{ttt} = -a^3 V_{xxx} + \mathcal{O}(k^2, h^2)$$

← what does V_{xxx} do to the solution?

$$V_t + aV_x = C V_{xxx} \rightarrow V = e^{i(\xi x - (a\xi + C\xi^3)t)}$$

$$\text{Ansatz: } e^{i(\xi x - \omega t)} \rightarrow V = e^{i\xi(x - (a + C\xi^2)t)}$$

$$V_t = -i\omega V$$

$$V_x = i\xi V$$

$$V_{xxx} = (i\xi)^3 V$$

$$-i\omega V + a i\xi V = C (i\xi)^3 V$$

$$\omega - a\xi = C\xi^3$$

$$\omega = a\xi + C\xi^3$$

$$\begin{cases} V(x,t) = \tilde{V}(x - (a + C\xi^2)t) \\ u(x,t) = u_0(x - at) \end{cases}$$

Each wavenumber ξ has a slightly different speed.
(numerical dispersion)